

which is enough to tank up 4,200 cars a day, 217 TPD of high-quality organic fertiliser, and 74 TPD of liquid CO₂, which is used in food processing and chemical industry. ATBL will invest about Rs 200 crore across phases to achieve name plate capacity.

Environmental Impact

The plant's operation aligns with the sustainability goals of the Adani Group. By utilising advanced anaerobic digestion technology, the plant converts organic materials into renewable biogas, significantly reducing greenhouse gas emissions and reliance on fossil fuels, besides aiding to the nation's fuel security and emission reduction goals. One of the most pressing issues in large swathes of north India is stubble burning, which significantly contributes to air pollution. By diverting agricultural waste to this facility, ATBL not only provides a solution to stubble burning but also generates eco-friendly bio-CNG and organic fertiliser, showcasing a perfect example of circular economy. Additionally, the plant contributes to soil health by enriching it with organic matter from the fertiliser.

Economic Benefits

The economic implications of the Barsana Plant are also profound. On the one hand, it provides income to farmers, cattle owners, as well as the gaushala who



can now monetise their organic waste. The CBG produced is a replacement for imported gas, thus leading our country towards energy self-sufficiency. As India is also a net fertiliser importer, the byproducts like organic fertiliser also aid the nation by cutting import requirements. The plant has opened up direct and indirect job opportunities in the predominantly agricultural region, fostering local economic growth.

Going Forward

The Barsana Biogas Plant is just the beginning of ATBL's ambitious CBG plans. The company is already working

on setting up waste-based CBG plants in Ahmedabad and Rajkot in Gujarat. These facilities will address the mounting challenges associated with municipal waste disposal and mitigate greenhouse gas emissions from landfills.

The expansion of ATBL's bio-CNG production facilities underscores the company's commitment to sustainable energy and environmental conservation. By leveraging advanced technology and innovative solutions, ATBL is poised to become a leader in India's renewable energy sector.

The Balancing Act

Sustainable farming aims to satisfy global food demands without depleting natural resources or causing environmental harm. Given the rising food demand, climate change, and shrinking resources, this farming model is more crucial than ever. It ensures a healthy balance between environmental stewardship, economic viability and social responsibility.

Using renewable energy sources such as solar, wind and biogas is essential for making agriculture more resilient and environmentally friendly. These renewable energies not only reduce the carbon footprint of farming but also provide a reliable and affordable energy supply. The innovation helps farmers tackle climate challenges and rising costs, enabling more food production with fewer resources.

